EXPT-02.
16-07-2026.

STUDY OF DIFFERENT TYPES OF NETWORK CABLES.

Aim:-

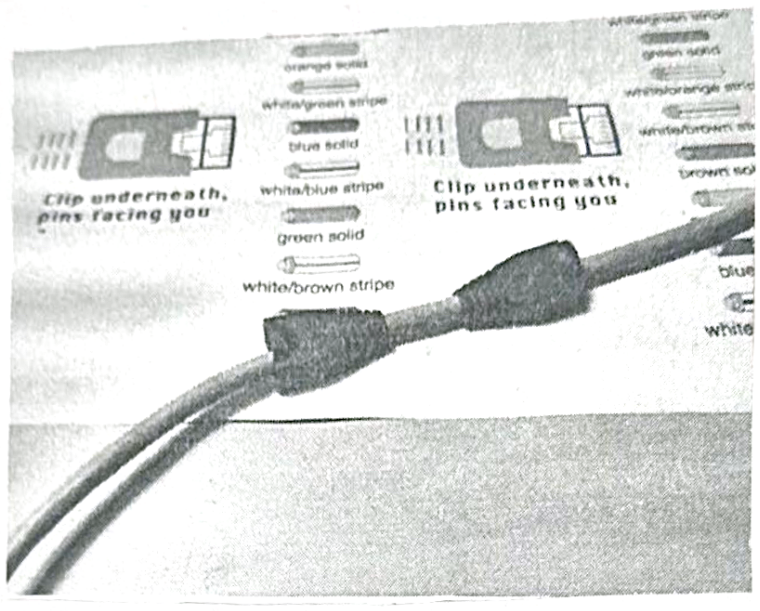
To study and understand different types of network cables, To make your own ethernet cross over cable/straight cable.

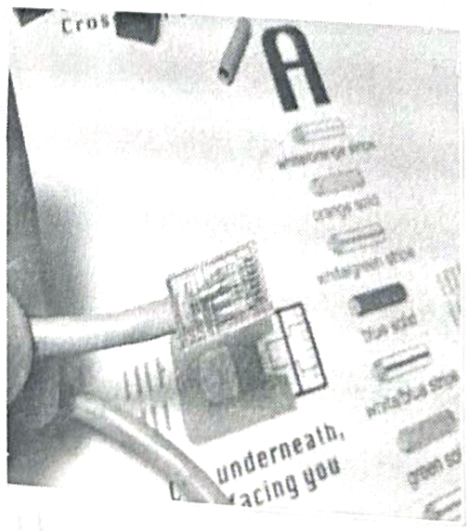
a) Understand different types of network cable.

Different types of cables used in networking are:

1. Unshielded Twisted Pairs (UTP) cables.
2. Shielded Twisted Pairs (STP) cables.
3. Coaxial cable
4. Fibre optic cable.

CABLE TYPE	CATEGORY	MAXIMUM DATA TRANSMISSION	ADVANTAGES / DISADVANTAGES	APPLICATION
UTP	Category 3	10 bps	ADVANTAGES: • Cheaper in cost • Easy to install as they have a smaller overall diameter. DISADVANTAGES:- • More prone to EMI and noise	10 Base-T Ethernet. Fast ethernet, Gigabit ethern
	Category 5	Up to 100 Mbps.		
	Category 5e	1 Gbps,		
STP.	Category 6, 6a	10Gbps.	ADVANTAGES: • Shielded, • Faster than UTP • Less susceptible to noise and interference. DISADVANTAGES: • Expensive. • Greater installation effort	Gigabit ethernet, 10G Ethernet (55 m). Widely used in data centres.
SSTP	Category 7.	10Gbps.		Gigabit Ethernet, 10G Ethernet, (100 m).


orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath,
pins facing you
Clip underneath,
pins facing you
green solid
brown

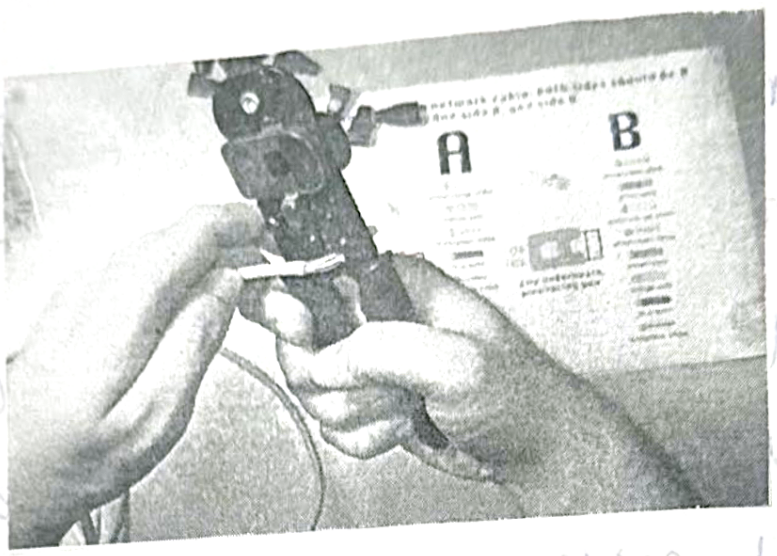

underneath,
acing you

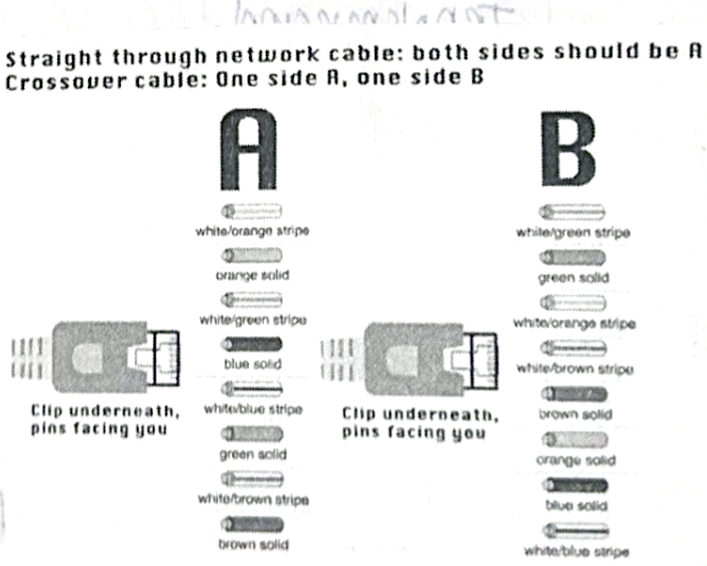
Coaxial cable.	RG-6 RG-59 RG-11	10-100 Mbps.	• High bandwith • Immune to Interference. • Low loss bandwidth. • Versatile. DISADVANTAGES: • Limited distance. • Cost. • Size is bulky.	Speed of signal is 500m Television network High speed internet connections.
Fibre optics cable.	Single mode Multi mode	100 Gbps.	ADVANTAGES: • High speed • High bandwidth • High security • Long distance DISADVANTAGES: • Expensive • Requires skilled installer.	Maximum distance of fibre optic cable is around 100 metres.

MAKE YOUR OWN ETHERNET CROSS-OVER CABLE/STRAIGHT CABLE:-

TOOLS AND PARTS NEEDED:-

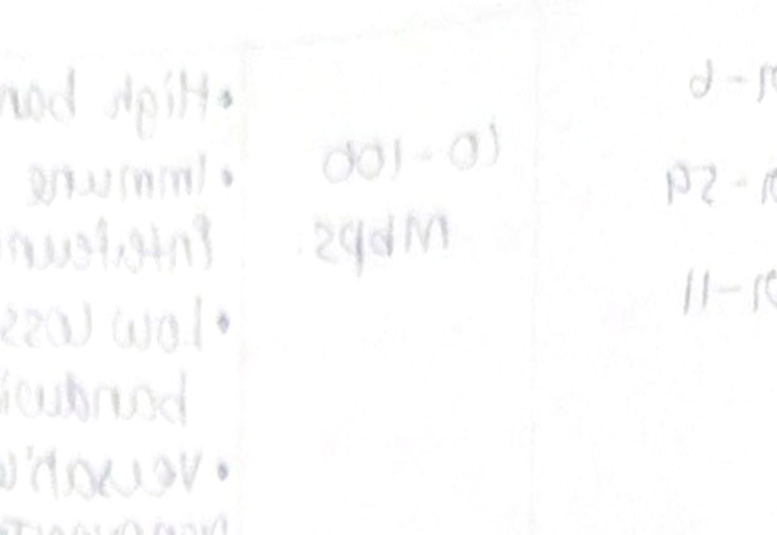
* Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.
* A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
* TWO RJ45 plugs.
* Optional two plug shields.


A
B


Straight through network cable: both sides should be A
Crossover cable: One side A, one side B
A
white/orange stripe
orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
brown solid
Clip underneath, pins facing you
B
white/green stripe
green solid
white/orange stripe
white/brown stripe
brown solid
orange solid
blue solid
white/blue stripe
Clip underneath, pins facing you

STEPS:-

1. To start construction of the device, begin by threading shields onto the cable.
2. Next, strip approximately 1.5cm of cable shieldi from both ends. The crimping tool has a round area to complete the task.
3. After, you will need to untangle the wires; there should be four "twisted pairs". Referencing ~~pairs~~ back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.
4. Strengthen the wires, trim them evenly and keep the order correct. Hold the RJ45 plug with clip away and gold pins facing you. Slide wires in carefully without disturbing the sequence.
5. Next, push the cable right in. The notch at the end of the plug needs to be just over t cable shielding and if it isn't, that mean that you stripped off too much shieldin Simply snip the cables back a little more
6. After the wires are securely sitting inside the plug, insert it into the crimping tool and push down. It should shape correctly, but pushing too hard can crack the fragile plastic plug.



Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

7. Lastly, repeat the other end using diagram B to make a crossover cable, using diagram A to make a straight through cable.

To test it, plug it in and attempt to connect two devices directly.

RESULT:-

Thus, different types of network cables were studied successfully, and straight through and cross over ethernet cables were prepared and tested successfully.